

Task: NAS - Necklace

Lab 1. Available memory: 128 MB.

Date: 02.11.2025

Description

Bajtazar built a robot that can construct precious necklaces. The beads used to produce the necklace are arranged on a moving one-way conveyor belt. Each bead is characterized by its value (which can also be negative).

In each step, the robot decides whether to pick the next element or skip it, and then moves to the right (it never moves back and never changes its decision).

Unfortunately, the robot is defective and breaks if it chooses K consecutive beads in a row. Interestingly, the robot works correctly in other situations. Bajtazar detected this fault and decided to improve the robot, but only succeeded partially.

During one full run of the belt, the robot can pull K consecutive beads in a row **at most once**. If it tries to do this twice (e.g., pulling $K + 1$ consecutive beads implies breaking the rule if strictly interpreted as strictly K , but the text implies repeating the action of taking a block of size K), it will break.

Professor Makary was asked to create an algorithm to determine which beads should be used to create a necklace to obtain the highest possible value during a single run.

Input

The first line of input contains two integers separated by a single space, n and K ($1 \leq n \leq 100\,000$, $1 \leq K \leq 10$), representing the number of beads and the robot's parameter K . The second line contains n integers from the interval $[-10^9, 10^9]$, which are the values of consecutive beads on the belt.

Output

The program should print one number, which is the maximum value of the beads the robot can collect.

Example

Input 1:

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7 3
7 9 9 6 5 5 5
```

Output 1:

```
35
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Input 2:

5 2
5 5 1 5 5

Output 2:

15